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SUPERSEDING
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MILITARY SPECIFICATION

BEEF STEW, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers thermostabilized beef stew in flexible pouches for use as a component of the Meal, Ready-To-Eat, Individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Thermostabililized Foods for the Armed Forces

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies may be obtained from the Office of Drinking Water Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Frozen Carrots
U.S. Standards for Grades of Frozen Peas
U.S. Standards for Grades of Tomato Paste

(Copies may be obtained from the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456 Washington, DC 20090-6456.)

* Institutional Meat Purchase Specifications

(Copies may be obtained from the Standardization and Review Branch, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Room 2649, South Building, Washington, DC 20090-6456.)

US Standards for Condition of Food Containers

(Copies may be obtained from Chairman, Condition of Food Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0610, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

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Meat and Poultry Inspection Regulations

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Copies should be obtained from the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Copies should be obtained from the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Copies should be obtained from the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

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AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Copies should be obtained from the American Dehydrated Onion and Garlic Association, 251 Kearney Street, No. 410, San Francisco, CA 94108.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.2).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

* 3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination or portion of recognizable primal rounds (shank off, heel out, and knuckles, peeled) and subprimal cuts of the round, shank-off (top round, bottom round (heel out), and knuckles (peeled)). Recognizable cuts are those which, when compared to Institutional Meat Purchase Specifications (IMPS) cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. A portion is a certified recognizable cut from which objectionable tissue, such as tendons or heavy connective tissue, has been removed. The beef shall be in the fresh-chilled state and shall be in excellent condition, i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class, grade and cut of meat, and typical of meat which has been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show no more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, the meat shall show no evidence of freezing, defrosting, or mishandling.

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3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, or heavy connective tissue. The boneless trimmed beef shall meet the limitations specified in tables I and II.

* 3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained at an internal temperature of 28° to 50°F (inclusive).

b. Holding in the fresh chilled state for not more than 4 days after initial certification is permitted provided that the beef is maintained at an internal temperature of 28° to 40°F (inclusive).

c. Holding in the frozen state for not more than 120 days after initial certification is permitted provided that the beef is:

- frozen to an internal temperature of 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence
- held after storage at an internal temperature not to exceed 40°F when further processing is resumed.

* 3.2.1.3 Beef, cooked, frozen. Cooked, frozen beef logs or dices shall be prepared from beef that complies with 3.2.1, 3.2.1.1, and 3.2.1.2. The beef shall be processed in accordance with 3.3.1, 3.3.1.1, and 3.3.1.2, as applicable. The cooked beef logs or dices shall be packaged and vacuum sealed in water-impermeable material having an oxygen permeability rate of not more than 10 cc of oxygen per square meter per 24 hours at 73°F and 0 percent relative humidity and shall be frozen to an internal temperature of 0°F or below within 72 hours. The initial freezing (in-storage) date and processing and packaging requirements shall be certified by a USDA Agricultural Marketing Service (AMS) Livestock Division Agent. The packaged, frozen, cooked beef logs or dices shall have been held at an internal temperature of 0°F or below for a period not to exceed 60 days prior to pouch filling (see 3.4).

3.2.2 Potatoes.

3.2.2.1 Potatoes, fresh. The potatoes shall be fresh, firm, clean, sound and of a white flesh variety suitable for canning. The maximum specific gravity for the potatoes shall be 1.075 with a reducing sugar content of not more than 2.0 percent on a dry weight basis.

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3.2.2.2 Potatoes, dehydro-frozen, diced. The dehydro-frozen potatoes shall be firm, possess a good color and flavor and be of a white flesh variety suitable for canning. The maximum specific gravity for the potatoes shall be 1.075 with a reducing sugar content of not more than 2.0 percent on a dry weight basis. The potatoes shall be of the most recent season's crop. The potatoes shall be mechanically diced to approximately 3/8 inch cubes.

3.2.3 Carrots.

3.2.3.1 Carrots, fresh. The carrots shall be fresh, firm, of good color, and of a variety suitable for canning.

3.2.3.2 Carrots, frozen, diced. The frozen diced carrots shall be U.S. Grade A of the U.S. Standard for Grades of Frozen Carrots and shall be of the most recent season's crop. The carrots shall be mechanically diced to approximately 3/8 by 3/8 by 3/8 inch dimensions.

3.2.4 Peas, green.

3.2.4.1 Peas, green, fresh. The peas shall be fresh, sound, clean, sweet green peas of a variety suitable for canning. The shelled peas shall be well-formed, bright, young, tender, and of such size as not to pass through a 9/32-inch sieve.

3.2.4.2 Peas, green, frozen. Frozen green peas shall be U.S. Grade A of the U.S. Standard for Grades of Frozen Peas and shall be of the most recent season's crop.

3.2.5 Starch, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized, modified maize food starch for use in thermostabilized foods and shall comply with MIL-STD-900. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The cooked, modified starch slurry shall be bland with essentially no cereal or starch taste.

3.2.6 Tomato paste. Tomato paste shall be U.S. Grade A of the U.S. Standards for Grades of Tomato Paste.

3.2.7 Margarine. Margarine shall conform to the FDA Standard of Identity for Margarine and shall possess a fine, pleasing flavor. The body shall be smooth, firm and homogeneous. The margarine shall possess a uniform medium yellow color and may be salted or unsalted.

3.2.8 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

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3.2.9 Sugar. Sugar shall be white refined, granulated cane or beet sugar, or a combination thereof, and shall comply with MIL-STD-900.

3.2.10 Onion powder. Onion powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.11 Vegetable protein, hydrolyzed. Hydrolyzed vegetable protein shall have been obtained by the hydrolysis of any one or any combination of corn, soybean or wheat proteins from which none of the monosodium glutamate has been extracted during processing. The product shall possess a characteristic color and flavor. Salt (chloride calculated as NaCl) shall not exceed 35 percent. The pH of a 2-percent solution shall be not less than 4.3 nor more than 7.0. Hydrolyzed protein shall be soluble in water at 90°F to the extent of not less than 5 grams in 100 milliliters.

3.2.12 Pepper, black, ground. The ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot biting pungent flavor and a light grey to speckled black-grey color. The black pepper shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.13 Celery, soluble. Soluble celery flavoring shall be prepared by properly blending and dispersing a minimum of 4 percent of total extractives of the dried fruit of Apium graveolens L. and soluble, dry, edible carrier. The flavoring shall possess a clean, celery-like aroma with a slightly bitter taste. The flavoring shall be free flowing, free from caking, and shall be yellowish-grey in color.

3.2.14 Color, caramel. Caramel color shall be a powder or liquid and shall comply with the Food Chemicals Codex.

3.2.15 Garlic powder. Garlic powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.16 Starch, filling and processing aid (see 6.3). The filling and processing aid starch shall be white, odorless, finely pulverized, modified waxy maize starch for use as a filling and processing aid in canned, retorted foods and shall comply with MIL-STD-900. During retort temperatures of 165° to 200°F, the starch shall form a clear, bland paste that retains its initial high viscosity under conditions of moderate shear and prolonged holding times. During the retort process the starch shall thin out and lend no appreciable viscosity to the final product.

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3.2.17 Sodium tripolyphosphate. Sodium tripolyphosphate shall comply with the Food Chemicals Codex.

3.2.18 Water. Water used for formulation, rehydration, blanching, ice-making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.19 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. Boned and trimmed beef which meets the requirements of 3.2.1.1 shall be further trimmed, if necessary, to assure compliance with finished product requirements and shall be processed in accordance with either 3.3.1.1 or 3.3.1.3.

3.3.1.1 Beef processing, log method. The beef shall be mechanically coarse ground through a grinder plate having kidney-shaped openings measuring not less than 2 inches in the shortest dimension. The grinder shall be equipped with a two-blade knife. The beef logs shall be formulated, prepared, and processed as follows:

Ingredient	Percent by weight
Beef	95.75
Ice or ice water	3.00
Salt	1.00
Sodium tripolyphosphate	0.25

a. The coarse ground (chunked) beef shall be mechanically vacuum mixed with the salt and sodium tripolyphosphate. The mixing shall continue until the mixture exhibits a sticky (tacky) consistency. Time from grinding to mixing shall not exceed 4 hours. Temperature of the meat mixture shall not exceed 40°F during this 4 hour time period.

b. The mixed meat shall immediately be mechanically tightly stuffed into casings of a size to accommodate the finished product meat dimension requirement. The stuffed beef logs shall be in the cooking process within 24 hours after being stuffed. The temperature of the logs shall be maintained at an internal temperature of 28° to 40°F during this 24 hour time period.

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c. The beef logs shall be cooked in a cookhouse (smokehouse without smoke) or by other commercially acceptable cook methods to provide a product meeting the finished product drained weight and other requirements.

d. Immediately after completion of the cooking process, the cooked beef logs shall be placed in an ice water bath or other commercially acceptable rapid cooling method. The cooled logs shall be held for not more than 48 hours at an internal temperature of 28° to 40°F prior to dicing or if stored in the frozen form until packaged and vacuum sealed.

e. The cooked beef may be stored in the frozen log form provided the logs are packaged and vacuum sealed in packaging material complying with the requirements specified in 3.2.1.3. The logs shall be frozen to at an internal temperature of 0°F or below within 72 hours of packaging. The packaged, frozen, cooked beef shall be maintained at an internal temperature of 0°F or below for a period not to exceed 60 days prior to pouch filling (see 3.4).

* 3.3.1.2 Dicing of cooked beef logs. The cooked beef logs shall be mechanically diced to yield uniform dices of approximately 5/8 by 5/8 by 5/8. The beef dices shall be screened for elimination of fines. The dices shall be held not longer than 48 hours at an internal temperature of 28° to 40°F until filled into the pouches or if stored in the frozen form until packaged and vacuum sealed (see 3.2.1.3). Dices prepared from previously frozen logs shall not be frozen.

3.3.1.3 Beef processing, dicing method. As an alternative to preparation of a log, the dices may be prepared from raw beef. The raw beef shall be diced and processed as follows:

a. The beef shall be diced to yield uniform dices of approximately 5/8 by 5/8 by 5/8 inch dimensions. The beef dices shall be screened for elimination of fines.

b. The raw beef dices shall be held for not more than 24 hours at an internal temperature of 28° to 40°F from time of dicing to beginning of the cooking process.

c. The beef dices shall be cooked in water, or by other commercially acceptable cooking method, until no raw portions are visible. The cooked beef dices shall be thoroughly drained.

d. The cooked beef dices shall be held for not more than 48 hours at an internal temperature of 28° to 40°F until filled into the pouches or, if stored in the frozen form, until packaged and vacuum sealed (see 3.2.1.3).

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3.3.2 Vegetable preparation.

3.3.2.1 Potato preparation.

* 3.3.2.1.1 Preparation of fresh potatoes. The fresh potatoes shall be washed, sorted, peeled and trimmed. The potatoes shall be mechanically diced to approximately 3/8 inch cubes. The dices shall be blanched sufficiently to prevent discoloration and to remove excess air. The blanched potatoes shall be immediately cooled to the initial temperature of the cooling water and thoroughly drained. The cooled, drained potato dices shall be handled in a manner to prevent discoloration and filled into the pouch within 4 hours after blanching.

* 3.3.2.1.2 Preparation of dehydrofrozen potatoes. Diced, dehydrofrozen potatoes shall be prepared in a manner so as to assure compliance with finished product requirements and shall be handled in a manner to prevent discoloration.

* 3.3.2.2 Carrot preparation. The carrots shall be washed, sorted, peeled, and trimmed. The carrots shall be mechanically diced to approximately 3/8 by 3/8 by 3/8 inch dimensions, and then blanched sufficiently to prevent discoloration and to remove excess air. The blanched carrot dices shall be immediately cooled to the initial temperature of the cooling water and thoroughly drained. The cooled, drained carrot dices shall be handled in a manner to prevent discoloration and filled into the pouch within 4 hours after blanching. When diced frozen carrots are used, they shall be handled in a manner to prevent discoloration, and shall be filled within 4 hours after removal from frozen storage.

3.3.3 Filling and processing aid starch solution preparation. When a filling and processing aid starch is used, a solution (slurry) shall be prepared with the filling and processing aid starch and cold water in a concentration adequate to hold ingredients in suspension for uniform distribution during mixing and pumping of product.

3.3.4 Sauce formulation and preparation. The sauce shall be formulated and prepared as follows:

Ingredients	Percent by weight
Water	79.85
Starch, modified, high opacity	5.88
Tomato paste (30% solids) 1/	5.88
Margarine	4.41
Salt 2/	1.50
Sugar	0.94
Onion powder	0.59
Vegetable protein, hydrolyzed	0.32
Pepper, black, ground	0.24

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Celery, soluble	0.18
Color, caramel 3/	0.18
Garlic powder	0.03

1/ The tomato paste shall be adjusted, as required, to produce a 30 percent tomato soluble solids level.

2/ The total amount of salt in the sauce formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6).

*3/ The total amount of caramel color shall be adjusted as necessary to produce a product that complies with the finished product color requirement (see 3.6).

NOTE: The following sauce preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.)

a. A slurry shall be made using the starch and part of the water.

b. The remaining ingredients shall be mixed together and heated to not less than 140°F.

c. The starch slurry shall be added and the final mixture heated to a temperature of 150° to 180°F and held in this temperature range for 5 minutes.

d. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The sauce shall be filled within 4 hours after cooking. As a final step before filling, a filling and processing aid starch solution (see 3.3.3) may be uniformly blended into the sauce in the amount required to facilitate uniform ingredient distribution in single-stage filling operations when dehydrofrozen potatoes are used. If a hot fill method is used, the prepared sauce shall be held at an internal temperature of 150° to 180°F at all times prior to filling into the pouches. If a cold fill method is used, the prepared sauce shall be cooled immediately and held at an internal temperature of 28° to 40°F at all times prior to filling into the pouches.

* 3.3.5 Product preparation. The diced beef, vegetables, and sauce shall be combined in the following approximate proportions, using either a single or multi-stage filling operation. The percent by weight of ingredients may be adjusted if necessary to assure compliance with finished product drained weight requirements.

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Ingredients	Percent by weight
Beef, diced	40.75
Sauce	33.80
Potatoes, diced 1/	15.00
Carrots, diced	7.50
Peas	2.95

1/ When dehydrofrozen potatoes are used, the 15.00 percent by weight shall consist of 7.50 percent by weight of dehydrofrozen potatoes and 7.50 percent by weight of water.

3.4 Pouch filling and sealing. Each pouch (see 5.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements specified for class 1 of MIL-P-44073.

b. A hot fill system is permitted if compatible with the single-stage or multi-stage filling and sealing method used.

c. The temperature of the diced beef, potatoes, carrots, and sauce at time of filling shall be as specified (see 3.3.1.2, 3.3.2 and 3.3.4) for the applicable filling process used.

d. Each filled and sealed pouch shall be in the retort process within 2 hours after sealing.

3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 1 of MIL-P-44073.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, glass, dirt, insect parts, hair, wood, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, moldy, rancid, sour, or stale.

c. There shall be no color foreign to the product.

d. Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, shall be not more than 0.35 ounce.

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e. The average net weight shall be not less than 8.0 ounces.

f. No individual pouch shall contain less than 7.5 ounces of product.

g. The average drained weight of beef shall be not less than 2.4 ounces.

h. No individual pouch shall contain less than 2.0 ounces of drained beef.

i. The average drained weight of vegetables (combined) shall be not less than 1.6 ounces.

j. No individual pouch shall contain less than 1.2 ounces of drained vegetables (combined).

* k. Of the 2.0 ounces drained weight of beef, not less than 1.5 ounces shall be whole intact beef dices.

l. Texture of the beef dices shall not be dry, rubbery, or mushy.

m. Texture of the vegetables shall not be mushy, hard, fibrous, or tough.

n. Sauce consistency shall be smooth, without lumps, and shall not be excessively thick or thin.

o. The sauce shall be a moderate brown color.

* p. The average fat content shall be not greater than 8.0 percent.

q. The fat content of the product in any individual pouch shall be not greater than 10.0 percent.

r. The salt content of the product in any individual pouch shall be not greater than 1.3 percent nor less than 0.5 percent.

s. The product shall show no evidence of excessive heating (materially darkened or scorched).

* t. No individual pouch shall contain a bone piece measuring 0.3 inch or more in any dimension.

3.6.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

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3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts or portions thereof. Any nonconforming beef shall be rejected.

4.5.1.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the AMS agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirements.

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The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or not being in excellent condition, the beef may be reworked by the contractor and reoffered for examination. For re-examination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be re-examined more than one time.

* TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (No. of sample units)	Defect categories Major AC	Major RE	Minor AC	Minor RE
500 or less	20	2	3	5	6
501 to 1200	32	3	4	7	8
1201 to 3200	50	5	6	10	11
3201 to 10000	80	7	8	14	15
10001 to 38000	125	10	11	21	22
Reinspection lots of 10001 to 38000	200	14	15	21	22

TABLE II. Boned and trimmed beef defects 1/ 2/

Category		Defect
Major	Minor	
101		Presence of popliteal, prescapular, prefemoral or any exposed lymph gland measuring 0.5 inch or more in any dimension.
102		Presence of blood clot measuring 1.0 inch or more in any dimension.
	201	Presence of blood clot measuring 0.5 inch or more but not less than 1.0 inch in any dimension.
103		Presence of bruise measuring 1.0 inch or more in any dimension.

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TABLE II. Boned and trimmed beef defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
	202	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
104		Presence of bone measuring 0.3 inch or more in any dimension.
105		Presence of cartilage measuring 0.5 inch or more in any dimension.
106		Presence of backstrap measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	203	Presence of backstrap measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
107		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more.
	204	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches.
	205	Presence of heavy connective tissue on lower edge of short plate or flank.
	206	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	207	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3.0 square inches or more.

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TABLE II. Boned and trimmed beef defects 1/ 2/ (cont'd)

Category		Defect
Major	Minor	
108		Presence of knuckle cover.
109		Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	208	Presence of kidney, pizzle eye, prepubic tendon, hanging tender or thymus gland measuring less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.
	210	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	211	Presence of dehydrated surface measuring 1.0 square inch or more.
	212	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more.
	213	Presence of exposed blood vessel measuring 1.0 inch or more in any dimension.
	214	Presence of cod, udder, kidney, or pelvic fat.

1/ Determination of wholesomeness and acceptability of product with respect to the presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal) shall be made by a Meat and Poultry Inspection Operations employee.

2/ Evidence of freezing or defrosting or product not in excellent condition shall be cause for rejection of the lot.

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4.5.1.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 7.5 ounces shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 8.0 ounces.

* 4.5.5 Product and drained weight inspection.

* 4.5.5.1 Product inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the thermoprocessing operation. The pouches shall be heated in 180° to 190°F water for 10 minutes, opened, and inspected. The sample size shall be as indicated by the sampling plan specified in table III. The contents shall be inspected for the defects indicated in table IV.

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TABLE III. Double sampling plan for product inspection 1/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

* TABLE IV. Product defects 1/ 2/

Category	Defect
Major	
101	Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, is more than 0.35 ounce.
102	Presence of a bone piece measuring 0.3 inch or more in any dimension.
103	Texture of beef dices is dry, rubbery, or mushy.
104	Texture of vegetables is mushy, hard, fibrous or tough.
105	Sauce not a moderate brown color.

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* TABLE IV. Product defects 1/ 2/ (cont'd)

Category	Defect
Major	
106	Sauce consistency lumpy or is excessively thick or thin.
107	Product shows evidence of excessive heating (materially darkened or scorched).

1/ The presence of foreign material (e.g. glass, dirt, insect part, hair, wood, metal), foreign odor or flavor (e.g., burnt, scorched, stale, sour, rancid, musty, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

* 4.5.5.2 Drained weight inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65° to 75°) after completion of the thermoprocessing operation. The sample size shall be as indicated by the double sampling plan specified in table V. The drained weight shall be determined and inspected for the defects listed in table VI.

TABLE V. Double sampling plan for drained weight inspection 1/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

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c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE VI. Drained weight defects 1/

Category	Defect
Major	
101	Drained weight of beef in a pouch is less than 2.0 ounces. 2/
102	Drained weight of vegetables (combined) in a pouch is less than 1.2 ounces. 3/
103	Of the 2.0 ounces drained weight of beef, less than 1.5 ounces are whole intact beef dices. 4/

1/ To determine drained weight, the pouch contents shall be poured into a flat bottom container. A minimum of three times the pouch's volume of 180° to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be gently agitated such as to liquify rendered fat and to remove the sauce without breaking the beef dices or vegetables. The contents shall then be poured into a U.S. Standard No. 7 (0.111 inch) sieve in a manner that will distribute the product over the sieve without breaking the beef dices or vegetables. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate 45 degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight of the beef and vegetables (combined) shall be determined individually and shall be reported to the nearest 0.1 ounce.

2/ If the sample average drained weight of beef is less than 2.4 ounces, the lot shall be rejected.

3/ If the sample average drained weight of vegetables (combined) is less than 1.6 ounces, the lot shall be rejected.

4/ For purposes of this examination, the definition of a whole intact beef dice is one that measures not less than 1/4 inch in each of two separate dimensions.

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4.5.6 Fat and salt content testing. The unopened sample pouches shall be warmed in a water bath to melt fat adhering to the inside of the pouches. The sample pouches shall be opened and the entire contents of each pouch shall be separately blended in a Waring blender or equivalent. The fat and salt content of the product from each pouch in the sample shall be determined in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Meat and Meat Products. Test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt requirement in 3.6 shall be classified as a major defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.5.7 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44073.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.2).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Starch, filling and processing aid. Shur-Fil ^{R} CS-500 starch, A.E. Staley Mfg. Co., Decatur, IL 62525 has been found to be satisfactory in holding ingredients in suspension for uniform distribution during mixing and pumping of product for single-stage filling operations.

* 6.4 Subject term (key word) listing.

Beef Stew
Meal, Ready-To-Eat
Ration
Thermostabilized

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6.5 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC
DP - SS

Preparing activity:

Army - GL

Project No. 8940-0649

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44059C	Beef Stew, Thermostabilized, For Meal, Ready-To-Eat

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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